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SUPERSEDING
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MILITARY SPECIFICATION

TRANSMITTING SET, RADIO AN/FRW-2 AND 2A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall transmit radio control signals to guided missiles and pilotless aircraft.

1.2 Classification - The Radio Transmitting Set shall be of two types and shall consist of the following items:

| Item | Type Designation | Part Of | Appl. Para. |
|---|---|---|---|
| Transmitter, Radio | T-560B/FRW-2 | AN/FRW-2 | 3.4.1 |
| Switch, R. F. Transmission Line | SA-431/U | AN/FRW-2 and 2A | 3.4.2 |
| Receiver, Radio | R-669A/URW | AN/FRW-2 and 2A | 3.4.3 |
| Decoder, Audio | KY-172/URW | AN/FRW-2 and 2A | 3.4.4 |
| Coder, Audio Frequency | KY-171/URW | AN/FRW-2 and 2A | 3.4.5 |
| Control, Coder | C-1669/URW | AN/FRW-2 | 3.4.6 |
| Amplifier, Audio Frequency | AM-1221A/URW | AN/FRW-2 and 2A | 3.4.7 |
| Transmitter, Radio | T-766/FRW-2A | AN/FRW-2A | 3.4.8 |

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-E-4682 | Electron Tubes and Transistors, Choice and Application of |
| MIL-W-5088 | Wiring; Aircraft, Installation of |

FSC 5820



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|---|---|
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-P-7788 | Plate, Plastic, Lighting |
| MIL-P-17555 | Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts |
| MIL-T-18303 | Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment |
| SAR-300 | Material Changes and Material Bulletins, Preparation of |
| MIL-E-16400 | General Specifications for Electronic Equipment Ship and Shore |

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration and the following factors shall govern:

(1) Parts and materials shall conform to either specification MIL-E-16400 or MIL-E-5400 as applicable.

(2) When the contractors can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as noninflammability, fungus resistance, nontoxicity, etc., and the meeting of environmental conditions as required by Specification MIL-E-16400 or MIL-E-5400 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall first be approved by the procuring agency.



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3.1.1 Nonstandard Part Approval and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in the Specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of either MIL-E-5400 or MIL-E-16400, and shall be submitted as nonstandard parts only when the material requirements are not met.

(2) Nonstandard parts as defined in either Specification MIL-E-5400 or MIL-E-16400 of an electrical nature must be approved before the preproduction model of the equipment is delivered to the procuring activity. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to, or in lieu of the Nonstandard Part Data Sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Power, A. C. Relays - Power, A. C. operated Relays employed in the AN/FRW-2 and 2A shall be of an approved type and approved for use in the equipment by the Bureau of Naval Weapons.

3.1.3 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.1.4 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of either MIL-E-5400 or MIL-E-16400 as applicable.

3.2 Design and Construction

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of either Specification MIL-E-5400 or MIL-E-16400 as applicable unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient (30° ± 10°C) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative |
| Input Power Voltage | 115 ± 1.0 V AC 60 cps<br>208 ± 2 V AC, 3 phase, 60 cps |

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in Specification MIL-E-16400 for Class 3 equipment except as modified herein.

(1) The equipment shall not be required to withstand any specified wind velocity.

(2) Ice loading need not be considered in the design of this equipment.

(3) The equipment shall not be required to meet the conditions of shock, vibration or inclination set forth in Specifications MIL-E-16400 or MIL-E-5400.



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(4) The equipment shall be operated continuously at full power output during all humidity tests.

3.2.4 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 1462 pounds.

3.2.5 Operational Stability - The equipment shall operate with specified performance for 500 hours, continuously or intermittently, without necessity for readjustment of any controls which are nonaccessible.

3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of 2000 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Cable Entrance - The input primary power and all interconnecting cables with the exception of the antenna cable shall enter through the base of the transmitter cabinet, which shall have provisions for mounting directly over a floor trench or for cable entrance from the rear at the floor level. Cable entrance to all external units shall be made at the rear of these units.

3.2.8 Interconnecting Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.9 Cables and Connectors - The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. Mating connectors which attach to all supplied equipment shall be included; however, the cables shall not be supplied as part of the equipment.

3.2.10 External Units - All portions of the equipment which are not located within the main cabinet shall be constructed as standard 19-inch relay rack units. The following external units may be removed for remote relocation:

(1) Control, Coder C-1669/URW

(2) Coder, Audio Frequency KY-171/URW

(3) Amplifier, Audio Frequency AM-1221A/URW

3.2.11 Input Power Requirements - The equipment shall operate satisfactorily from the following power sources:

(1) AC Power (Single Phase)

(a) 115 Volt 60 cps

| | |
|---|---|
| Operating Voltage Limits | 103.5 to 126.5 volts |
| Operating Frequency Limits | 55 to 65 cps |
| Power (1Ø) | 230 va |



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(2) AC Power (Three Phase)

(a) 208 Volt 60 cps

| | |
|---|---|
| Operating Voltage Limits | 198 to 228 volts |
| Operating Frequency Limits | 55 to 65 cps |
| *Power (3Ø) | 5 kva |

*The maximum unbalance of theload shall not exceed 50% between phases. The current required in any line shall not exceed 15 amperes.

3.2.12 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses or other protective devices shall be held to a minimum.

3.2.13 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified, but shall not automatically resume normal operation when the input voltage falls below the dropout potentials of the large power contactors.

3.2.14 Standby Operating Provisions - The transmitter shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.2.15 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400 or MIL-E-16400 as applicable. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.16 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.17 Interchangeability - The interchangeability requirements as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units and replaceable component parts of this equipment with the latest model of any previously produced equipments. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the procuring activity for that purpose. In the event that the requirements of MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.18 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of specification MIL-I-6181 with the following exceptions:

(1) Radiated Interference of a Harmonic nature shall be less than 1,000 microvolts between 270 and 800 megacycles, and less than 2,500 microvolts at frequencies of 800 megacycles and higher when measured as outlined in MIL-I-6181.

(2) Antenna conducted Interference - No Harmonic or spurious frequency shall exceed two (2) per cent of the unmodulated carrier power when measured as outlined in MIL-I-6181.

3.2.19 Safety - The design shall be such as to provide maximum convenience and safety to personnel installing, operating, and interchanging a complete equipment assembly or component part thereof. Satisfactory provisions shall be made to prevent personnel from accidentally coming in contact with voltage or binding posts. When cases and seals are removed for maintenance, protection shall be provided from voltages in excess of 500 volts.

3.2.20 Tuning and Adjustment - Every consideration shall be given to make adjustment of the transmitter for maximum efficiency a quick, simple operation involving a minimum of controls. Tuning shall be a local operation.

3.2.21 Special Circuitry - This equipment shall incorporate the following special circuitry.



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3.2.21.1 Monitoring Provisions - A monitor receiver shall be provided so that the radio-frequency power output is sampled and the audio content isolated. Provisions shall be made so that a high impedance recorder can be connected to one of two jacks which will afford a sample of modulation signal input or monitor receiver output.

3.2.21.2 Station Guardian -A station guardian shall be provided so that when the equipment is used as a continuous carrier device a decrease of transmitter power output or an increase of reflected power from the antenna will transfer operation to a standby equipment. A common antenna will be used for the two transmitters. The transfer mechanism (SA-431/U) shall provide that complete transfer shall occur in less than 1.0 second.

3.2.21.3 Input Impedance - Audio input impedances of 110 ohms balanced and 560 ohms unbalanced shall be provided on a selectable basis.

3.2.22 Warmup Time -A warmup time to permit the equipment to reach stable operation is allowable, but such time shall not exceed 15 minutes.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified. All system performance requirements shall be checked by using a receiver equivalent to or better than the R-423/ARW-56, which includes provisions for one-half megacycle channel spacing, as the completing link of the system.

3.3.2 Frequency Characteristics - AN/FRW-2: The radio frequency carrier shall be within the range of 406 to 549 megacycles inclusive and shall consist of 144 fixed channels spaced at one megacycle intervals on the integral megacycle. AN/FRW-2A: The radio frequency carrier shall be within the range of 406 to 549.5 megacycles inclusive and shall consist of 288 fixed channels spaced at one-half megacycle intervals on the integral and integral plus one-half megacycle points. (See paragraph 3.4.3.9.)

3.3.3 Modulation Characteristics - AN/FRW-2: The equipment shall provide for frequency modulation, capable of a maximum of 300 kilocycle deviation each side of the center frequency, when a modulating signal of between 300 cycles and 100 kilocycles is supplied from a source external to the transmitter. AN/FRW-2A: The equipment shall provide for frequency modulation, capable of a maximum of 300 kilocycle deviation each side of the center frequency, when a modulation signal of between 600 cycles and 100 kilocycles is supplied from a source external to the transmitter.

3.3.4 Keyed Carrier and Continuous Carrier Provisions - A switch shall be provided, the function of which shall be to select transmitter operation with a continuous carrier or with a keyed carrier which is keyed by the external modulation. The output power rating shall be with continuous carrier.

3.3.5 Tolerances - AN/FRW-2: The radio frequency carrier center frequency shall be maintained within ±0.01 per cent of the frequency to which the equipment is tuned. The audio modulating tones shall be held within ±1 per cent of the specified frequency. AN/FRW-2A: The radio frequency carrier center frequency shall be maintained within ±21 kilocycles of the frequency to which the equipment is tuned. The audio modulating tones shall be held within ±1 per cent of the specified frequency.

3.3.6 Transmitter Power Output - The transmitter shall supply a minimum of 500 watts of radio frequency power when connected to a nonreactive load of 52 ohms.

3.3.7 Audio Frequency Response - The frequency response shall, when compared to a 10-kilocycle reference, not vary beyond the following limits:

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(1) AM-1221A/URW Bypassed

(a) Compressor operating: +1 to -2 db from 7 to 80 kc.

(b) AN/FRW-2, compressor Bypassed: ±1 db from 1 to 25 kc, and +1 to -3 db from 0.3 to 100 kc.

(c) AN/FRW-2A, compressor Bypassed: ±1 db from 1 to 25 kc, and +1 to -3 db from 0.6 to 100 kc.

(2) AM-1221A/URW operating.

(a) Compressor operating: +2 to -4 db from 7 to 80 kc.

(b) AN/FRW-2, Compressor Bypassed: +2 to -6 db from 0.3 to 100 kc.

(c) AN/FRW-2A, Compressor Bypassed: +2 to -6 db from 0.6 to 100 kc.

3.3.8 Distortion - The over-all amplitude distortion in the system shall not exceed the following limits:

AN/FRW-2

(1) AM-1221A/URW Bypassed.

(a) Compressor Operating: 3.5% with an applied signal frequency between 0.3 and 1 kc, and 2.5% between 1 and 100 kilocycles

(b) Compressor Bypassed: 3% with an applied signal frequency between 0.3 and 1 kc, and 2% between 1 and 100 kilocycles.

(2) AM-1221A/URW Operating:

(a) Compressor Operating: 4.0% with an applied signal frequency between 0.3 and 100 kilocycles.

(b) Compressor Bypassed: 3.5% with an applied signal frequency between 0.3 and 100 kilocycles.

AN/FRW-2A

(1) AM-1221A/URW Bypassed.

(a) Compressor Operating: 3.5% with an applied signal frequency between 0.6 and 100 kilocycles.

(b) Compressor Bypassed: 3% with an applied signal frequency between 0.6 and 100 kilocycles.

(2) AM-1221A/URW Operating:

(a) Compressor Operating: 4% with an applied signal frequency between 0.6 and 100 kilocycles.

(b) Compressor Bypassed: 3.5% with an applied signal frequency between 0.6 and 100 kilocycles.

3.3.9 Polarity - The modulation polarity of the AN/FRW-2 and AN/FRW-2A transmitters shall be the same.

3.4 Detail Requirements -

3.4.1 Transmitter, Radio T-560B/FRW-2 - The radio transmitter shall meet the following requirements:

3.4.1.1 Function - The transmitter shall generate and amplify a radio-frequency carrier, and provide a method of frequency modulation for said carrier when an audio signal is applied. When coupled to a proper antenna the transmitter shall produce a signal complying with all limitations outlined as performance data in this specification.



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3.4.1.2 Form Factor and Mounting - The transmitter shall be housed in a cabinet designed for floor mounting. The overall dimensions of the cabinet shall not exceed 72 inches high, 28 inches wide, and 28-1/2 inches deep (exclusive of handles on doors).

3.4.1.3 Contents of Unit - The transmitter shall contain all the necessary circuitry to perform the function given. The following subassemblies shall be included:

| | |
|---|---|
| Compressor | Driver Servo Amplifier |
| Modulator | Mechanical Tuning |
| Low Frequency IF. | Power Amplifier |
| High Frequency Osc. | P. A. Screen Supply |
| High Frequency IF. | Station Guardian |
| Resonator-Tripler | Control Panel |

In addition to the above assemblies the transmitter shall contain a Relay Panel, Fuse Panel, Meter Panel, Interconnecting Cabling, Monitor Panel, and Power Supplies.

3.4.1.4 Weight - The weight of the transmitter including the cabinet, but exclusive of the Radio Receiver R-669A/URW, Audio Decoder KY-172/URW and the R. F. Switch SA-431/U for which space shall be provided in the same cabinet, shall not exceed 1200 pounds.

3.4.1.5 Electrical Connections - A terminal strip shall be mounted on the lower left front corner of the cabinet for connection of the three-phase input power. A monitor panel shall be mounted on the lower left front corner of the transmitter which shall support the following receptacles:

(1) Two type UG-422/U coaxial connectors for 110 ohm balanced line input.

(2) Two type UG-290/U coaxial connectors for line transformer output.

(3) Two type AN3102A-16S-1P, 7-pin connectors for interconnections for dual installation.

(4) Two type UG-290/U coaxial connectors for compressor input.

(5) Two type UG-290/U coaxial connectors for decoder input.

(6) Two type UG-290/U coaxial connectors for receiver output.

(7) One type AN3100A-28-15S (0°), 35 pin connector for decoder output.

(8) One type AN3100A-28-21P (0°), 37 pin connector for decoder output.

(9) One type AN3100A-18-8P (0°), 8 pin connector for spare connections.

3.4.1.5.1 Antenna Connections - The radio frequency output of the transmitter unit shall terminate in a receptacle mounted at the top of the cabinet to mate with type UG-154/U coaxial cable connector to which RG-17/U or RG-18/U cable may be connected.

3.4.1.6 Controls - All controls necessary for normal operation of the transmitter shall be located on a control panel at a convenient height for adjustment. This panel shall be installed at the front of the transmitter and must be accessible when the cabinet door is closed. The controls located on the control panel shall be in accordance with the following table:



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| Control Name | Identification Lettering | Type of Control |
|---|---|---|
| Channel Switch | CHANNEL SWITCH | Concentric knob rotary switches |
| Main Power Switch | ON-OFF MAIN POWER | Dual push-button switch |
| Exciter Power Switch | ON-OFF EXCITER | Dual push-button switch |
| Driver Power Switch | ON-OFF DRIVER | Dual push-button switch |
| Power Amplifier Tuning Power Switch | ON-OFF POWER AMPLIFIER TUNE | Dual push-button switch |
| Power Amplifier Operating Power Switch | ON-OFF POWER AMPLIFIER OPERATE | Dual push-button switch |
| Manual Automatic Switch | MANUAL DIRECT, REFLECTED POWER; AUTOMATIC DIRECT, REFLECTED POWER | Four position rotary switch |
| Power Amplifier Grid Current Meter Switch | SCREEN CURRENT, GRID CURRENT | Two position rotary switch |
| Exciter-Driver Current Meter Switch | EXCITER GRID, DRIVER GRID, DRIVER PLATE | Three position rotary switch |
| Driver Fine Tuning Control | DRIVER TRIM | Rotary potentiometer |
| Voltage Meter Switch | LINE VOLTAGE, P. A. FILAMENT VOLTAGE | Two position rotary switch |
| Power Amplifier Fine Tuning Control | P. A. TRIM | Rotary potentiometer |
| Power Amplifier Screen Voltage Control | P. A. SCREEN VOLTAGE | Rotary potentiometer |
| Keyed Carrier Switch | KEYED CARRIER, NORMAL | Two-position rotary switch |

All other controls shall be easily accessible upon opening cabinet doors or sliding shelves.

3.4.1.7 Power Requirements - The transmitter shall operate from a power source supplying a voltage of 208 volts, three phase, 60 cycles per second, and shall require no more than 5 kva and shall be capable of normal operation with supply voltage variations of +10, -5 per cent for transient or sustained periods of four hours maximum. Primary transformer taps shall be provided to permit operation from the following supply voltages: 200, 208, 218 and 230 volts per phase.

3.4.1.8 Accessory Power - The transmitter shall supply power to the following:

R-669A/URW, Radio Receiver; 115 volts, single-phase, 60 cycles per second, 2 amperes.

SA-431/U, R. F. Transmission Line Switch; 115 volts, single phase, 60 cycles per second, 1.5 amperes.



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3.4.1.9 Operational Simplicity - Every consideration shall be given to design which will make adjustment of the transmitter a quick, simple operation involving a minimum number of controls.

3.4.1.10 Tuning - Tuning shall be accomplished automatically when the filament and exciter power switches are turned on and the appropriate channel selected. Channel selection to cover the radio frequency band of 406 to 549 megacycles shall be accomplished by the use of concentric knobs. Adjustment of fine tuning controls shall be a manual operation.

3.4.1.11 Frequency Characteristics - The transmitter shall provide for operation on each of 144 channels spaced at one-megacycle intervals on the integral megacycle within the range of 406 to 549 megacycles inclusive.

3.4.1.12 Frequency Control - The frequency shall be controlled by not more than 25 quartz crystals. A complete set of crystals necessary to cover the entire frequency band shall be provided.

3.4.1.13 Carrier Frequency Drift - The center frequency shall not drift more than ±0.01% of the selected frequency when the equipment is operated for a period of 24 hours, continuously or intermittently, without adjustment under any of the conditions specified herein.

3.4.1.14 Radio Frequency Power Output - The radio frequency power output delivered to a 52 ohm noninductive load shall be not less than 500 watts under all conditions specified.

3.4.1.15 Modulator - A modulator shall be included with the transmitter capable of producing an output carrier frequency deviation from ± 30 kilocycles to ± 300 kilocycles at any modulating frequency from 300 cycles to 100 kilocycles. The modulator shall be provided with an adjustment ( on the modulator ) to permit obtaining from ± 30 kilocycles to ± 300 kilocycles carrier frequency deviation for applied sine wave signal voltages from 0.85 to 10.0 volts rms.

3.4.1.16 Compressor - A compressor circuit with a resistive input impedance of approximately 560 ohms shall be provided which will accept, through an RG-58/U coaxial cable, the output of the KY-171/URW Audio Frequency Coder. The compressor shall supply a voltage to the modulator which maintains substantially constant deviation when going from one to six tones. For a 7.5 kilocycle tone at an input level of 6.0 volts rms, the deviation shall be ± 300 kc. When the amplitude of this tone is varied over a range of 6.0 to 0.8 volt rms, the deviation shall not vary more than plus 0 and minus 10 per cent. A circuit shall be provided for bypassing the compressor stage.

3.4.1.17 Keyed Carrier Provisions - An internal switch shall be provided to select operation of the transmitter as a continuous carrier or a keyed carrier device. For keyed carrier operation, when the modulation is applied, the carrier shall go on in 15 milliseconds or less and go off within the range of 100 to 175 milliseconds when the modulation is removed.

3.4.1.18 Harmonic and Spurious Components - Harmonic and spurious frequency components shall be controlled within the limits specified in paragraph 3.2.18(2).

3.4.1.19 Station Guardian - A station guardian shall be provided so that a change in the transmitter output will transfer the operation to a standby transmitter. The station guardian shall be adjustable by means of locking screw driver adjustments to allow any change of power between the limits of -1 and -6 db to initiate the transfer. The Power Output Meter will indicate reflected power as well as incident power in the antenna circuit.

3.4.1.20 Meters - The following functions shall be metered separately to provide a means of monitoring the transmitter operation without the use of circuit selector switches: Carrier Power Output, Carrier-Frequency Deviation, Line Voltage and Power Amplifier Plate



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Current. Selector switches and/or additional meters shall be provided to indicate the following for tuning and adjustment: Exciter Grid Current, Driver Grid Current, Driver Plate Current, Power Amplifier Grid Current, Power Amplifier Screen Current, Power Amplifier Filament Voltage, and Reflected Power from the antenna.

3.4.2 Switch, R. F. Transmission Line SA-431/U - The R. F. transmission line switch unit shall meet the following requirements:

3.4.2.1 Function - The transmission line switch shall transfer a common antenna from an operating transmitter to a standby transmitter upon failure of the operating unit. The station guardian shall operate this switch.

3.4.2.2 Form Factor and Mounting - The switch shall be flange mounted. The radio frequency portion shall be located on one side of the flange and the power frequency portion on the other side. Maximum overall dimensions of the case shall be as follows: 9-3/8 inches high, 7-1/4 inches long and 5 inches wide.

3.4.2.3 Contents of Unit - The switch shall contain a rotary switch wafer, a motor and a set of motor driven radio frequency contacts.

3.4.2.4 Weight - The weight of the unit shall not exceed 14 pounds.

3.4.2.5 Electrical Connections - Three coaxial radio frequency connectors shall be included of a type which will mate type UG-154/U connectors. They shall provide input connections for the radio frequency output of the two transmitters and an output connection for the antenna. One AN3102C-16S-8P connector shall be supplied for connection of motor power and travel limit indication circuitry.

3.4.2.6 Power Requirements - The actuating power required shall be 115 volts, single phase, 60 cycles per second at 1.5 amperes rms.

3.4.2.7 Power Handling Capability - The rf transmission line switch shall have a carrying capacity not less than 1 kilowatt of power with the switch contacts in either stationary position. During switching operations the radio frequency contacts shall be shorted to ground.

3.4.2.8 Frequency Characteristics - The switch shall give satisfactory operation at any frequency within the range of 400 to 600 megacycles.

3.4.2.9 Voltage Standing Wave Ratio - The voltage standing wave ratio shall be less than 1.5 to 1 at any frequency from 400 to 600 megacycles.

3.4.2.10 Insertion Loss - Insertion loss shall be less than 0.3 db from 400 to 600 megacycles.

3.4.2.11 Characteristic Impedance - The transmission line switch shall be designed for an impedance of 52 ohms.

3.4.2.12 Switching Time - Switching time shall not exceed 0.60 second.

3.4.2.13 Contact Resistance - Contact resistance shall not exceed 50 milliohms.

3.4.2.14 Contact Arrangement - The radio frequency contacts shall be of the single pole, double- throw type.

3.4.3 Receiver, Radio R-669A/URW - The radio receiver shall meet the following requirements:

3.4.3.1 Function - The receiver shall receive a sample of the carrier produced in the transmitter for purposes of separating and monitoring the modulation thereon. The receiver



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shall amplify the detected signal to a voltage suitable for operation of the Audio Decoder KY-172/URW.

3.4.3.2 Form Factor and Mounting - The receiver shall be constructed for standard 19 inch relay rack mounting, except that rails shall be provided to support the weight upon a shelf instead of hanging by means of the front panel. The maximum overall dimensions of the chassis and panel shall be 19-1/32 inches wide, 8-3/4 inches high and 18-9/16 inches deep. The overall dimensions including cable bend radii and plug removal clearance shall not exceed 19-1/32 inches wide, 8-3/4 inches high, and 21-1/2 inches deep.

3.4.3.3 Contents of Unit - The receiver unit shall consist of the following plug-in assemblies:

First Injection Oscillator

First Intermediate Frequency Amplifier

Second Intermediate Frequency Amplifier

Audio Amplifier

Power Supply

Junction Box

3.4.3.4 Weight - The weight of the receiver shall not exceed 70 pounds.

3.4.3.5 Electrical Connections - The following external receptacles shall be provided on the receiver:

(1) One type UG-291/U coaxial connector for the radio frequency input connection.

(2) One type UG-290/U coaxial connector for the audio output connect

(3) One 2 pin connector type AN3102A-12S-3P for input power connection.

(4) One 6 pin connector type AN3102A-18-12S for connection of the direct current power supply to the audio decoder.

3.4.3.6 Controls - The frequency selector controls shall be located on the front panel of the receiver. An opening shall be provided in the transmitter cabinet so that receiver tuning may be accomplished without interruption of the system operation. The controls shall be marked with the operating frequency of the receiver in megacycles, and shall consist of two knobs for rotating selector switches. All adjustments not used in normal operation shall be easily accessible upon sliding the receiver forward in its rack.

3.4.3.7 Power Requirements - The receiver shall meet all specified requirements when operating from a source of 103.5 to 126.5 volts, single phase, 55 to 65 cycles per second. The receiver shall require a maximum of 2.2 amperes.

3.4.3.8 Accessory Power - The following power shall be available from the monitor receiver power supply to operate the audio decoder:

+280 volts direct current at 65 milliamperes



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-18 volts direct current for bias

6.3 volts alternating current at 7 amperes.

3.4.3.9 Frequency Characteristics - The receiver shall operate on each of 144 channels spaced at one megacycle intervals on the integral megacycle within the range of 406 to 549 megacycles inclusive.

3.4.3.10 Receiver Sensitivity - The receiver shall have a sensitivity of 0.05 volt with a minimum signal plus noise to noise ratio of 30 db measured at the receiver output with a deviation of ±300 kilocycles

3.4.3.11 Frequency Stability - The design of the receiving equipment shall be such that a center frequency drift of ±0.01% from the selected frequency under any service conditions shall not impair the operating performance (i.e., distortion or phase linearity).

3.4.3.12 Local Oscillators - The local oscillators shall be frequency controlled by not less than one (1) or more than two (2) crystals for each channel frequency. Twenty-five (25) different crystals (total) shall cover the entire frequency range and shall be supplied with each equipment. The oscillators shall be designed to beat with the incoming signal so as to provide reception at the integral megacycle points throughout the tuning range. All crystals shall be of standard Navy type.

3.4.3.13 Bandwidth - The receiver shall be capable of accepting deviations from zero to ±300 kilocycles. The overall 6 db bandwidth shall not exceed 2.0 megacycles nor be less than 1.4 megacycles.

3.4.3.14 Receiver Fidelity - The receiver shall have a response characteristic such that the output will not vary more than ±1 db when the modulating frequencies are between 1 and 25 kilocycles, not more than +1 db or -3 db between 300 cycles per second and 100 kilocycles when compared to the output at a reference frequency of 10 kilocycles.

3.4 3.16 Blocking - The receiver shall not block when signals of 2 volts or less are applied.

3.4.3.17 Audio Output - The audio output has been designed for an external load impedance approximately of 560 ohms with a resistive path to ground. For one tone at ±300 kilocycles deviation the output voltage shall have an adjustment range from 1.4 to 7 volts peak. Six audio tones may be received simultaneously. The output voltage shall be controlled by a locking type adjustment.

3.4.4 Decoder, Audio KY-172/URW - The audio decoder shall meet the following requirements:

3.4.4.1 Function - The decoder unit shall accept the output of the R-669A/URW or other suitable receiver, separate it into its twenty component signal-channel frequencies (corresponding to those generated by the coder unit) and provide "On-Off" switching for external circuits associated with the frequency selective channels.

3.4.4.2 Form Factor and Mounting - The maximum dimensions of the decoder shall not exceed 19-1/32 inches wide by 5-1/4 inches high by 16-1/4 inches deep. The decoder is constructed for standard rack mounting. The overall dimensions including cable bend radii and plug-removal clearance shall not exceed 19-1/32 inches wide, 5-1/4 inches high and 19 inches deep.

3.4.4.3 Contents of Unit - The decoder shall contain all the circuitry necessary to perform its stated function, and shall consist of the following:

Audio Amplifier and Cathode Follower



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20 Filter Circuits

20 Amplifier-relay actuation stages

20 Relays

3.4.4.4 Weight - The total weight of the unit shall not exceed 38 pounds.

3.4.4.5 Electrical Connections - The following electrical receptacles shall be provided on the rear of the chassis:

(1) A coaxial signal input connector type UG-447/U.

(2) Two relay output connectors, one 37 pin and one 35 pin of types AN3102A-28-21P and AN3102A-28-15P respectively.

(3) A 6 pin power input connector type AN3102A-18-12P.

(4) An 8 pin spare connector type AN3102A-18-8P.

(5) Two binding-post test terminals for test metering purposes.

3.4.4.6 Controls - All controls and adjustments are to be internal. They shall be readily accessible when the equipment is removed from the rack.

3.4.4 7 Power Requirements - The operating power shall be supplied by the R-669A/URW Radio Receiver power supply as follows:

(1) Direct current requirements -

238 to 322 volts at 70 milliamperes maximum

15 to 24 volts negative bias.

(2) Alternating current requirements -

5.5 to 7.1 volts rms 55 to 65 cps at 8.3 amperes maximum

3.4.4.8 Input - The decoder shall present an input impedance of approximately 560 ohms resistive. When tested in a complete system consisting of Audio Frequency Coder KY-171/URW, Radio Transmitter T-560B/FRW-2 or T-766/FRW-2A, and Radio Receiver R-669A/URW, with the receiver output set at level of 2 volts rms (for the 7.50 kilocycle channel alone), the decoder shall operate satisfactorily (all channels operating and no false operation of channels) with from one to six tones actuated.

3.4.4.9 Audio Channels and Relays - Each of the 20 frequency selective channels shall have a hermetically sealed relay for opening and closing an external circuit when the particular channel is energized by a signal of the proper frequency. The frequencies shall be the same as those listed in paragraph 3.4.5.12. The relay for controlling the external circuit shall perform the function of a double-pole, double-throw switch which is normally de-energized in the absence of a signal in its frequency selective circuit. The time delay between application of a signal to the decoder and closure of the contacts shall be kept to a minimum and shall not exceed 15 milliseconds. Similarly the release time shall not exceed 15 milliseconds. The relay contacts shall be rated at 28 V dc at 2 amperes for a resistive load.

3.4.5 Coder, Audio Frequency KY-171/URW - The audio frequency coder shall meet the following requirements:

3.4.5.1 Function - The coder shall generate 20 control signals of frequencies between 7500 and 75,000 cycles per second. These signals shall be suitable for modulating the



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carrier of the transmitter.

3.4.5.2 Form Factor and Mounting - The coder shall be constructed for standard relay rack mounting with maximum dimensions of 19-1/32 inches wide, 7 inches high and 15-1/2 inches deep.

3.4.5.3 Contents of Unit - The audio frequency coder shall contain all the electrical components necessary to perform its stated function. The following circuitry shall be included:

20 Oscillator Circuits

Cathode Follower

Power Supply

Voltage Regulator

3.4.5.4 Weight - The weight of the coder shall not exceed 35 pounds.

3.4.5.5 Electrical Connections - The following electrical receptacles shall be provided on the rear of the unit.

(1) A 2 pin power input connector type AN3102A-12S-3P.

(2) Two 19 pin keying control connectors, one type AN3102A-22-14P and one AN3102A-22-14S.

(3) A coaxial signal output connector type UG-447/U.

3.4.5.6 Controls - A single power "On-Off" switch shall be located on the front panel. All other controls and adjustments shall be internal and easily accessible when the coder is removed from the rack.

3.4.5.7 Power Requirements - The coder shall operate satisfactorily under all specified conditions from a source of 103.5 to 126.5 volts rms, single phase, 55 to 65 cycles per second. A maximum rms current of 0.60 ampere shall be required.

3.4.5.8 Accessory Power - The coder unit shall supply voltages of 108 volts regulated direct current and 6.3 volts, single phase, 60 cycles per second at 0.6 ampere for operation of the Coder Control C-1669/URW.

3.4.5.9 Output - The coder unit shall provide an output signal between 0.85 and 1.5 volts rms per tone across an external resistive load of approximately 560 ohms. A circuit shall be incorporated capable of mixing from one to six control signals to form the output signal of this unit. The output voltage may vary depending on the number of signal channels simultaneously operated.

3.4.5.10 Interaction - The design of the audio channels of the coder unit is such that signals of any one, or up to six channels will operate only the correct relays in the decoder unit.

3.4.5.11 Coder - Decoder - Under standard conditions, the coder and decoder shall be capable of operating one channel at a time, when connected together by means of a 52-ohm line.



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3.4.5.12 Frequency Characteristics - The audio frequency channels shall be as follows:

| Audio Tone No. | Frequency (Kc) | Audio Tone No. | Frequency (Kc) |
|---|---|---|---|
| 1 | 7.50 | 11 | 25.01 |
| 2 | 8.46 | 12 | 28.21 |
| 3 | 9.54 | 13 | 31.83 |
| 4 | 10.76 | 14 | 35.90 |
| 5 | 12.14 | 15 | 40.49 |
| 6 | 13.70 | 16 | 45.68 |
| 7 | 15.45 | 17 | 51.52 |
| 8 | 17.43 | 18 | 58.12 |
| 9 | 19.66 | 19 | 65.56 |
| 10 | 22.17 | 20 | 73.95 |

3.4.5.13 Frequency Stability - Frequency drift of signal-channel frequencies shall not exceed ±1 per cent under any of the operating conditions herein specified.

3.4.6 Control, Coder C-1669/URW - The coder control shall meet the following requirements:

3.4.6.1 Function - When operated manually the coder control shall key the proper signal-channel of the coder. Each switch shall have independent operation and may be used independently or in any combination up to six simultaneously.

3.4.6.2 Form Factor and Mounting - The coder control shall be constructed for standard 19 inch relay rack mounting with maximum dimensions of 19-1/32 inches wide, 7 inches high, and 6-1/4 inches deep.

3.4.6.3 Contents of Unit - The coder control shall contain the necessary wiring for all interconnections between 2 receptacles, 13 switches, and 6 indicating lights.

3.4.6.4 Weight - The total weight of the equipment shall not exceed 4.5 pounds.

3.4.6.5 Electrical Connections - Two multicontact receptacles shall be mounted on a side bracket for connection to the coder oscillator circuits. The receptacles shall be as follows:

One 19 pin type AN3102C-22-14P and one 14 pin type AN3102C-20-27P.



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3.4.6.6 Controls - The following controls shall be provided and identified on the face of the unit:

| Control Name | Identification Lettering | Type of Control |
|---|---|---|
| Brake Switch | BRAKES, ON-OFF | 2 position-center off-toggle switch |
| Flap Switch | FLAPS, UP-DOWN | 2 position-center off-toggle switch |
| Throttle Switch | THROTTLE, ADV. - RET. | 2 position-center off-toggle switch |
| Rudder Switch | RUDDER, L. -R. | 2 position-center off-toggle switch |
| Wheel Switch | WHEELS, UP-DOWN | 2 position-center off-toggle switch |
| Control Stick | U. -D., L. -R. | Special multiple switch |
| Centering Switch | | Push switch |
| Auxiliary Switch | | 2 position-center off-toggle switch |
| Auxiliary Switch | | 2 position-center off-toggle switch |
| Auxiliary Switch | | 2 position-center off-toggle switch |

3.4.6.7 Power Requirements - The switching voltage supplied to the coder control shall be 108 volts regulated d-c for controlling circuits in the coder. An alternating current voltage of 5.67 to 6.93 volts rms single phase, 55 to 65 cycles per second at a maximum of 0.7 ampere rms shall be required for indicating lights.

3.4.7 Amplifier, Audio Frequency AM-1221A/URW - The audio frequency amplifier shall meet the following requirements:

3.4.7.1 Function - The audio-frequency amplifier shall provide impedance matching and power gain for supplying line losses when the coder control, coder, and this unit are installed at a remote location and coupled to the transmitter by means of a 110-ohm balanced transmission line.

3.4.7.2 Form Factor and Mounting - The amplifier unit shall be designed for standard 19 inch relay rack mounting and shall have maximum dimensions of 19-1/32 inches wide, 5-1/4 inches high and 9-3/4 inches deep.

3.4.7.3 Contents of Unit - The audio-frequency amplifier shall contain all the electronic components necessary to perform its stated function. The following circuitry shall be included:

Phase Inverter

Power Amplifier

Power Supply



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Voltage Regulator

Output Transformer

3.4.7.4 Weight - The total weight of the unit shall not exceed 30 pounds.

3.4.7.5 Electrical Connections - The following electrical receptacles shall be located on the rear of the chassis:

(1) A 5 pin power connector type AN3102A-14S-5P.

(2) A twin coaxial audio output connector type UG-422/U.

(3) A coaxial audio input connector type UG-447/U.

3.4.7.6 Controls - A power "On-Off" toggle switch and an input signal amplitude control potentiometer shall be located on the front panel of the amplifier.

3.4.7.7 Power Requirements - The audio amplifier unit shall meet all the requirements herein specified when operated from a source of 103.5 to 126.5 volts, single phase, 55-65 cycles per second, and shall require a maximum of 1.2 amperes.

3.4.7.8 Input - The amplifier has been designed to present an impedance of approximately 500 ohms resistive to an audio signal of 300 cycles to 100 kilocycles. The input level may be variable over the range of 0.85 to 1.5 volts rms per tone. The amplifier shall be capable of accepting from 1 to 6 audio tones applied simultaneously.

3.4.7.9 Harmonic Distortion - Harmonic distortion introduced by the audio-frequency amplifier shall not exceed 2.0% from 0.3 to 100 kilocycles per second.

3.4.7.10 Noise Level - The noise level shall be held to a minimum and shall under no condition exceed a level of 50 db below the normal signal level.

3.4.7.11 Frequency Response - The frequency shall, when compared with a 10 kilocycle reference, not vary greater than +1 or -3 db from 0.3 to 100 kilocycles.

3.4.7.12 Power Gain - The audio frequency amplifier shall provide a minimum power gain of 6 db with the gain control at a maximum under all conditions specified.

3.4.8 Transmitter, Radio T-766/FRW-2A -

3.4.8.1 Function - The transmitter shall generate and amplify a radio-frequency carrier, and provide a method of frequency modulation for said carrier when an audio signal is applied. When coupled to a proper antenna the transmitter shall produce a signal complying with all limitations outlined as performance data in this specification. The Transmitter, Radio T-766/FRW-2A provides for an improvement over the T-500B/FRW-2 in that a new Modulator filfills the requirement for narrow and wide Band frequency modulation of the R.F. carrier of Transmitting Set, Radio AN/FRW-2A. The new modulator also provides operation of the AN/FRW-2A on 288 filaments spaced one-half (1/2) megacycle apart between 406 and 549.5 megacycles. (See paragraph 3.4.3.9).

3.4.8.2 Form Factor and Mounting - The transmitter shall be housed in a cabinet designed for floor mounting. The over-all dimensions of the cabinet shall not exceed 72 inches high, 28 inches wide, and 28-1/2 inches deep (exclusive of handles on doors).



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3.4.8.3 Contents of Unit - The transmitter shall contain all the necessary circuitry to perform the function given. The following subassemblies shall be included:

| | |
|---|---|
| Compressor | Driver Servo Amplifier |
| Modulator | Mechanical Tuning |
| Low Frequency IF | Power Amplifier |
| High Frequency Osc. | P.A. Screen Supply |
| High Frequency IF. | Station Guardian |
| Resonator-Tripler | Control Panel |

In addition to the above assemblies the transmitter shall contain a Relay Panel, Fuse Panel, Meter Panel, Interconnecting Cabling, Monitor Panel, and Power Supplies.

3.4.8.4 Weight - The weight of the transmitter including the cabinet, but exclusive of the Radio Receiver R-669A/URW, Audio Decoder KY-172/URW and the R. F. Switch SA-431/U for which space shall be provided in the same cabinet, shall not exceed 1225 pounds.

3.4.8.5 Electrical Connections - A terminal strip shall be mounted on the lower left front corner of the cabinet for connection of the three-phase input power. A monitor panel shall be mounted on the lower left front corner of the transmitter which shall support the following receptacles:

(1) Two type UG-422/U coaxial connectors for 110-ohm balanced line input.

(2) Two type UG-290/U coaxial connectors for line transformer output.

(3) Two type AN3102A-16S-1P, 7-pin connectors for interconnections for dual installation.

(4) Two type UG-290/U coaxial connectors for compressor input.

(5) Two type UG-290/U coaxial connectors for decoder input.

(6) Two type UG-290/U coaxial connectors for receiver output.

(7) One type AN3100A-28-15S (0°), 35-pin connector for decoder output.

(8) One type AN3100A-28-21P (0°), 37-pin connector for decoder output.

(9) One type AN3100A-18-8P (0°), 8-pin connector for spare connections.

3.4.8.6 Antenna Connections - The radio frequency output of the transmitter unit shall terminate in a receptacle mounted at the top of the cabinet to mate with type UG-154/U coaxial cable connector to which RG-17/U or RG-18/U cable may be connected.



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3.4.8.7 Controls - All controls necessary for normal operation of the transmitter shall be located on a control panel at a convenient height for adjustment. This panel shall be installed at the front of the transmitter and must be accessible when the cabinet door is closed. The controls located on the control panel shall be in accordance with the following table:

| Control Name | Identification Lettering | Type of Control |
|---|---|---|
| Channel Switch | CHANNEL SWITCH | Concentric knob rotary switches |
| Main Power Switch | ON-OFF MAIN POWER | Dual push-button switch |
| Exciter Power Switch | ON-OFF EXCITER | Dual push-button switch |
| Driver Power Switch | ON-OFF DRIVER | Dual push-button switch |
| Power Amplifier Tuning Power Switch | ON-OFF POWER AMPLIFIER TUNE | Dual push-button switch |
| Power Amplifier Operating Power Switch | ON-OFF POWER AMPLIFIER OPERATE | Dual push-button switch |
| Manual Automatic Switch | MANUAL DIRECT, REFLECTED POWER; AUTOMATIC DIRECT, REFLECTED POWER | Four position rotary switch |
| Power Amplifier Grid Current Meter Switch | SCREEN CURRENT, GRID CURRENT | Two position rotary switch |
| Exciter-Driver Current Meter Switch | EXCITER GRID, DRIVER GRID, DRIVER PLATE | Three position rotary switch |
| Driver Fine Tuning Control | DRIVER TRIM | Rotary potentiometer |
| Voltage Meter Switch | LINE VOLTAGE, P.A. FILAMENT VOLTAGE | Two position rotary switch |
| Power Amplifier Fine Tuning Control | P.A. TRIM | Rotary potentiometer |
| Power Amplifier Screen Voltage Control | P.A. SCREEN VOLTAGE | Variable transformer |
| Keyed Carrier Switch | KEYED CARRIER, NORMAL | Two-position rotary switch |

All other controls shall be easily accessible upon opening cabinet doors or sliding shelves.

3.4.8.8 Power Requirements - The transmitter shall operate from a power source supplying a voltage of 208 volts, three phase, 60 cycles per second, and shall require no more than 5 kva and shall be capable of normal operation with supply voltage variations of +10, -5 per cent for transient or sustained periods of four hours maximum. Primary transformer taps shall be provided to permit operation from the following supply voltages: 200, 208, 218 and 230 volts per phase.



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3.4.8.9 Accessory Power - The transmitter shall supply power to the following:

R-669A/URW, Radio Receiver 115 volts, single-phase, 60 cycles per second, 2 amperes.

SA-431/U, R. F. Transmission Line Switch 115 volts, single phase, 60 cycles per second, 1.5 amperes.

3.4.8.10 Operational Simplicity - Every consideration shall be given to design which will make adjustment of the transmitter a quick, simple operation involving a minimum number of controls.

3.4.8.11 Tuning - Tuning shall be accomplished automatically when the filament and exciter power switches are turned on and the appropriate channel selected. Channel selection to cover the radio frequency band of 406 to 549.5 megacycles shall be accomplished by the use of concentric knobs. Adjustment of fine tuning controls shall be a manual operation.

3.4.8.12 Frequency Characteristics - The transmitter shall provide for operation on each of 288 channels spaced at one-half megacycle intervals on the integral and integral plus one-half megacycle points within the range of 406 to 549.5 megacycles inclusive.

3.4.8.13 Frequency Control - The frequency shall be controlled by not more than 27 quartz crystals. A complete set of crystals necessary to cover the entire frequency band shall be provided.

3.4.8.14 Carrier Frequency Drift - The center frequency shall not drift more than ±21 kilocycles of the selected frequency when the equipment is operated for a period of 24 hours, continuously or intermittently, without adjustment under any of the conditions specified herein.

3.4.8.15 Radio Frequency Power Output - The radio frequency power output delivered to a 52-ohm noninductive load shall be not less than 500 watts under all conditions specified.

3.4.8.16 Modulator - A modulator shall be included with the transmitter capable of producing an output carrier frequency deviation from ± 30 kilocycles to ± 300 kilocycles at any modulating frequency from 600 cycles to 100 kilocycles. The modulator shall be provided with an adjustment (on the modulator) to permit obtaining from ± 30 kilocycles to ± 300 kilocycles carrier frequency deviation for applied sine wave signal voltages from 0.85 to 10.0 volts rms.

3.4.8.17 Compressor - A compressor circuit with a resistive input impedance of approximately 560 ohms shall be provided which will accept, through an RG-58/U coaxial cable, the output of KY-171/URW Audio Frequency Coder. The compressor shall supply a voltage to the modulator which maintains substantially constant deviation when going from one to six tones. For a 7.5 kilocycle tone at an input level of 6.0 volts rms, the deviation shall be ± 300 kc. When the amplitude of this tone is varied over a range of 6.0 to 0.8 volt rms, the deviation shall not vary more than plus 0 and minus 10 per cent. A circuit shall be provided for by-passing the compressor stage.

3.4.8.18 Keyed Carrier Provisions - An internal switch shall be provided to select operation of the transmitter as a continuous carrier or a keyed carrier device. For keyed carrier operation, when the modulation is applied, the carrier shall go on in 15 milli-seconds or less and go off within 175 milliseconds when the modulation is removed.

3.4.8.19 Station Guardian - A station guardian shall be provided so that a change in the transmitter output will transfer the operation to a standby transmitter. The station guardian shall be adjustable by means of locking screwdriver adjustments to allow any change of power between the limits of -1 and -6 db to initiate the transfer. The Power Output Meter will indicate reflected power as well as incident power in the antenna circuit.



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3.4.8.20 Meters - The following functions shall be metered separately to provide a means of monitoring the transmitter operation without the use of circuit selector switches: Carrier Power Output, Carrier-Frequency Deviation, Line Voltage and Power Amplifier Plate Current. Selector switches and/or additional meters shall be provided to indicate the following for tuning and adjustment: Exciter Grid Current, Driver Grid Current, Driver Plate Current, Power Amplifier Grid Current, Power Amplifier Screen Current, Power Amplifier Filament Voltage, and Reflected Power from the antenna.

3.4.8.21 Deviation Monitor - Circuitry shall be provided as part of the modulator assembly to permit monitoring the modulation deviations over the range of ± 30 to ± 300 kilocycles. A deviation indicator shall be provided and shall be calibrated in three (3) ranges: zero (0) to 300 kc. deviation, zero (0) to 150 kc. deviation, and zero (0) to 50 kc. deviation, over the full-scale deflection of the indicator. This indicator shall be similar in size and appearance to the indicator furnished in the T-560B/FRW-2, and shall mount in that indicator's place.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-E-16400 and interference tests in accordance with Specification MIL-I-6181.

4.2.3.1 Shock; Bench Handling - Chassis or chassis and front panel type assembly shall be removed from its enclosure, as for servicing, placed in a suitable position for servicing on a solid 2 inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(1) The assembly shall be tilted upward through an angle of 30 degrees, using one edge of the assembly as an axis, and permitted to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as an axis, for a total of four drops.

(2) Repeat (1) with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly could be placed practicably for servicing.



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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment. It may not be resubmitted as the production sample model.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample tests the equipment shall be reworked to the contractor for any necessary rework. The equipment shall then be resubmitted for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.



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4.4.1.1 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration (10 minutes) | Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the government inspector for sampling tests:

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 10 | 1 |
| Next 50 | 1 |
| Next 75 | 1 |
| Next 100 | 1 |
| | One for each additional 200 or fraction thereof |

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 30 hours.



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4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.



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4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 25 | 1 |
| Next 175 | 1 |
| Next 300 | 1 |
| | 1 for each additional 500 or fraction thereof |

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

| | |
|---|---|
| Temperature | Normal room |
| Altitude | Normal ground (0 - 5000 ft.) |
| Humidity | Room ambient |
| A.C. Voltage | 115 ± 5 volts single phase 60 cps |
| A.C. Voltage | 208 ± 5 Volts three phase 60 cps |

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.



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4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

## 6. NOTES

6.1 Use - The AN/FRW-2 and 2A shall generate audio tones which frequency modulate a uhf carrier. The monitor system shall detect and isolate the audio tones for operation of frequency sensitive circuitry which will actuate relays. These equipments shall be for installation in permanent buildings.



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6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1 volt ac and 208 ± 2 volts ac. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.



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